

Spring 2026 ASTR 4140/5140 Research Methods in Astrophysics

Project #1 -- Complete by Jan 22, 2026.

The only work which should be handed in is your modified Jupiter notebook + plot and modified PS1_bounce.py program from the last exercise below. Please comment your code and upload it to Collab. Many thanks to Marla Geha for providing the tutorials for this project.

Introduction to Programming: The goal of this section is to set up your programming environment. Prior programming experience in Python is not required, but I expected that you have some familiarity with general programming. We will start with an introduction to the command line and Unix commands (which you can skip if this is already familiar).

1. The Command Line: The command line is the starting point for most astrophysics research computing. From the command line, you can move/copy files or startup other software. To open a command line terminal:

If you are using a Mac computer: Go to the Applications -> Utilities -> Terminal. Once Terminal opens use the pull down menu to open a window Shell -> New Window -> Basic, or type `cmd+N`.

If you are using a Windows machine: We suggest using PowerShell to access the terminal which should already be installed. If this doesn't work, try installing "winbash" to access the terminal: <http://sourceforge.net/projects/win-bash/files/shell-complete/latest/>. If windbash does not work, an alternative is cygwin.

2. Basic Unix Commands: The command line accepts commands in a language called "bash" for UNIX. We will denote UNIX commands with '>' (further below, python commands are denoted as '>>>'). A full UNIX tutorial can be found under 'Resources' on the Collab site. Read through this tutorial and execute the following tasks.

List the contents of your home directory (`> ls`). Create a new directory under your home directory called 'ASTR4140' (`> mkdir ASTR4140`). From Collab/Resources, downloaded the file named PS1_bounce.py into your new directory. View the contents of this file using the 'more' command (`> more PS1_bounce.py`). This is a simple python script which we will use below.

3. Text Editing: When writing programs, we need to edit files without the forced formatting of programs such as Word or Powerpoint. There are a number of options for text editing. Most astronomers usually use Sublime, Emacs or vi for Mac. You can download Emacs for free (<http://ftp.gnu.org/pub/gnu/emacs/>), and as I recommend in the Unix Tutorial, I like Aquamacs. vi comes preinstalled on Macs. Sublime is available at: <http://www.sublimetext.com/>

Sublime is available for Mac or Window users. Windows users can also try Note- pad+++. Inside your new class directory, create a text file called "text.txt". Type the sentence "This

is a test file.” at the top of the file. Exit the file and use the UNIX command `>more text.txt` to verify you have created the file correctly. Alternatively you can use Jupyter notebooks.

4. Displaying images: We will use a stand alone program to display astronomical image called DS9 (stands for Deep Space 9, and yes, that’s a Star Trek reference). First download the appropriate version of the code from:

<http://hea-www.harvard.edu/RD/ds9/site/Download.html>

For Mac users, click on the downloaded disk image (dmg) and move SAOImage DS9 file into your Applications folder. For Windows users, run the executable file to install. Opening the application will bring up a new window. Click on file -> open and import the file from the class Files PS1_image.fits (you will need to download it first). Play with the scaling of the image by clicking scale -> linear/log/etc. Move your mouse over the image and notice the numbers in the upper left panel. Click on zoom and zoom into the image so that you can see individual pixels. Click on color and change the color scale. Note that the actual numbers in this image (the numerical value of each pixel) don’t change. You can change the contrast with your mouse (Mac users, press command while dragging your mouse on the image).

5. Installing Python: Python programming will be our primary research tool throughout class. Python is a widely used language, and has been adopted by the majority of astronomers. Basic python is already installed on Mac machines, but you will need additional libraries for our work. Windows machine will need to install both python and additional libraries. There are multiple installations for scientific python libraries. I suggest the Anaconda distribution available at:

<https://www.anaconda.com/download/>

The distribution is free, but you will need to input a valid email address. Download and follow instructions for installing this software.

To test if your installation is successful, open a terminal and type ‘ipython’. We suggest using either the ipython environment or Jupyter notebooks for your work. In ipython, you should see the command: In [1]: To quit out of python type ‘exit’ or type control-D twice.

6. Learning Python: There are many online resources for learning python. I’ve linked to videos and online courses. If you have no prior programming experience, I suggest working through the Google Python Class lectures (~2 hours of video):

Watch [Google Python Class 1.1](#). Do first exercise ([string1.py](#)).

- Watch [Google Python Class 1.2](#). Do second exercise ([list1.py](#)).
- Watch [Google Python Class 1.3](#). Note minute 9:30 when the Google search engine is explained in two lines of code. Do third exercise ([wordcount.py](#)).
- A good short tutorial is <https://learnpythonthehardway.org/python3/> which is free up through Exercise #8.
- Another good tutorial is <http://www.learnpython.org/>.

7. How cool is Python?: We want to give you a preview of what python can do. Try running the program you downloaded called ‘PS1_bounce.py’ by typing:

```
> ipython --pylab
IN[1]: run PS1_bounce.py
IN[2]: show()
IN[3]: run PS1_bounce.py
```

If at first you get error messages, don't worry! Try understanding these messages and debug the error, but if it makes no sense, talk to Prof. K or the TA. You will know if the program is working.

8. Python Notebooks: You have run the above program using a python script and the command line. It is possible to do great research this way. However, we prefer Jupyter Notebook: <https://jupyter.org>, which is an interactive computational environment. This allows you to export your notebook (plus plots).

On Macs, you can start a notebook by typing 'jupyter-notebook' in the command line (this will open a browser window). On Windows: Open up the Windows home navigator in the bottom left corner of the screen in the taskbar. There should be a "Recently Added" section that includes an application called "Anaconda Navigator." If the app doesn't appear here, search for it in the taskbar. Click on this app, and when it opens, the first screen it takes you to has an option on the left hand side of the screen to launch Jupyter Notebooks. If you click on this, it will open the notebook directly into your browser.

Go through the notebook of best practices made originally by Connor McClellan, and edited by Nick (best_practices.ipynb). Also look at notebook_help.ipynb. These will be an excellent resource moving forward. Several useful links and files are available under the Module "Python_resources". Please familiarize yourself with these as part of this assignment.

More Useful Python Links:

- [Style Guide](#) for python code.
- Jupyter NBs: A Crash Course in Python for Scientists: <https://nbviewer.jupyter.org/gist/rpmuller/5920182>

9. Make the code your own: There are three parts to this problem set. First, ensure that you can load in the data file "planets.csv", as instructed, and second, modify the plot created in the notebook PS1_notebook.ipynb (available under Files) as requested in the comments of the file (there are 4 requests, including adding axes title to the plot). To learn more about plotting see [a tutorial here](#), [sample plots here](#) or [a wordy but pretty good discussion on plotting](#).

Please save your notebook as a .ipynb file (File > Download As > ipython notebook) and upload this file to Collab.

Second, modify the python code PS1_bounce.py in at least 3 significant ways (e.g., change color, shape, number of objects or something more creative). Document your changes as comments in the code itself and upload to Collab. If you need inspiration, examples can be found here: <https://python4astronomers.github.io/contest/bounce.html>

Extra credit will be awarded to any unique or sophisticated modifications.